



Welcome to the class of
Web Information Retrieval !

Min ZHANG (张敏)
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Course basic information

Schedule

- Time: **13:30 - 15:55, Tues.** (with two breaks)
 - 13:30 - 14:15; 5 minutes' break;
 - 14:20 - 15:05; **5 minutes' break;**
 - **15:10 - 15:55.**
- QA: at WeChat Group (online), or Open Hour (offline) 
- THU Web Learning platform
 - <http://learn.tsinghua.edu.cn>
 - Homework assignment and submission
 - Slides sharing after class



清华大学
Tsinghua University

网络学堂
WEB LEARNING

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Instructor



Instructor: Min ZHANG (张敏)

- Tenured Full Professor
- IR group, DCST, Tsinghua Uni.; AI institute, THU.
- Office: FIT 1-506A
- Email: z-m@tsinghua.edu.cn
- <http://www.thuir.cn/group/~mzhang>

TA: Vadim GRIGOREV (樊紫龙)

- Email: Gr.Vadimvit@gmail.com
- Office: FIT 1-506

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A little bit about Min (1)



Experiences and Services

- Bachelor: 1999, PhD: 2003, in DCST, THU
- Now Full Professor in THUIR group, CST Dept.
- Visited NUS, DFKI, Kyoto Uni., City Uni. of HK, MSRA
- [ACM Distinguished Member](#), [ACM SIGIR Academy](#), [TOIS Editor-in-Chief \(2020-now\)](#)

Research Interests

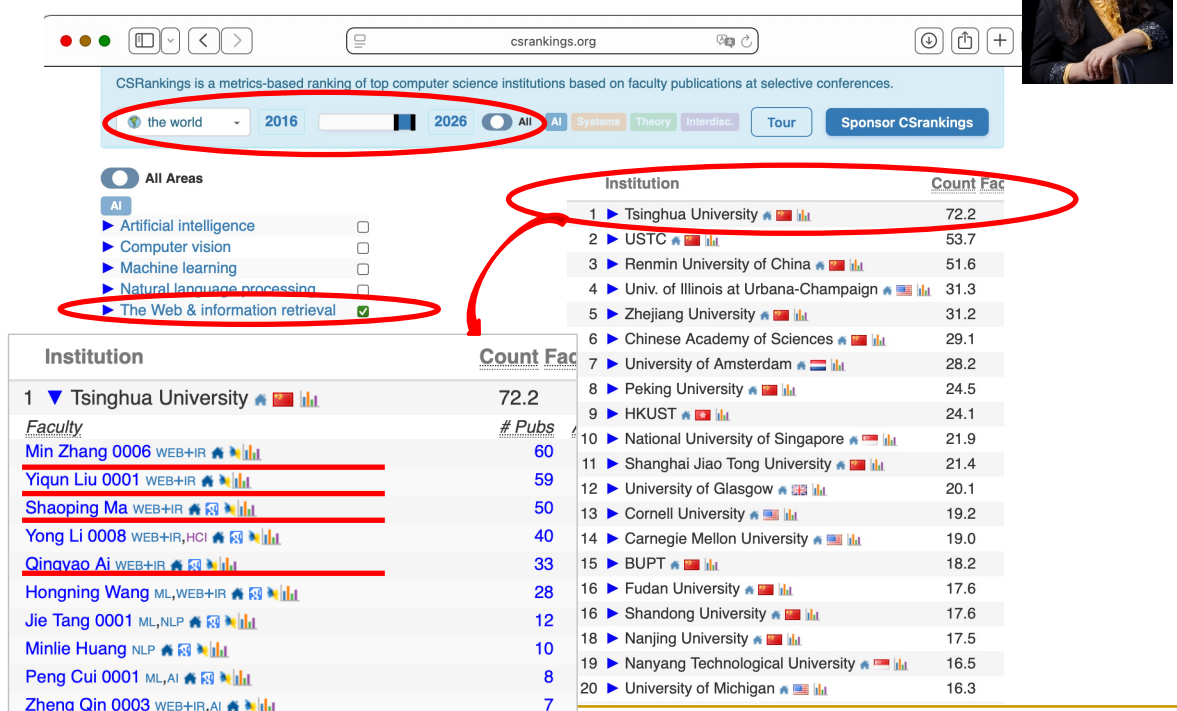
- Information Retrieval & Recommendation, User Modeling, Machine learning applications

Achievements and Impacts

- Top conferences/journals publications, >18000 citations (Feb. 2026)
- Multiple Top Performances in TREC and NTCIR since 2002.
- General or PC Chair: [ACM SIGIR'26](#), MM'25, CHIIR'24, RecSys'23, CIKM'23, ICTIR'20 short, EVIA'19, SIGIR'19 tutorial, WSDM'19 workshop, SIGIR'18 short, WSDM'17.....
- [SIGIR24 Test of Time Award](#), [EMNLP24 Best resource](#), [WSDM22 best paper...](#)
- [One of the top ranked World-wide IR researchers](#)

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A little bit about Min (2)



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A little bit about Min (3)



Course Experiences

- **Introduction to Machine Learning (2003 – Now)**
 - **The first** Machine Learning course in CST Dept., Tsinghua Univ.
- Advanced Topics in Information Retrieval (2008- Now)
 - Graduate Students
- Web and IR (for Advanced Computing Program, 2011 - Now)
 - **Credit Course in THU-CMU Double Master Degree Program in CS**

Awards

- Excellent Young Faculty Teaching Award, Tsinghua University, 2008
- **Outstanding Teacher of Computer Science in Chinese Universities**, China, 2018.
- Excellent Online Teaching Award, Tsinghua University, 2020
- **IBM Global Faculty Award** (2021)
- **Changjiang Scholar** (2022)
- Okawa Foundation (2022)

Collaborations with industry:

- Kuaishou, Tencent, TikTok, China Mobile, Huawei, Meituan, Microsoft, Alibaba, VMware, Youdao
.....

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Course Arrangements (Tentative):Part I

- **Part I: The first 8 weeks, Lectures by the teacher**
 - From the **Historical** and **Systematic** Perspective
 - IR systems/architectures/models **Evolution**
 - Evaluation
 - Recommendation
 - Data issues matter
 - What is the **new generation IR**?
 - Three strategic reports
 - US SIGIR(July, 2024)
 - China CCIR (Jan., 2025)
 - Australia SWIRL (Feb., 2025)



SIGIR

CCIR



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Course Arrangements (Tentative):Part I

- Part I: The first 8 weeks, Lectures by the teacher
 - On selected topics
- + By students: “Tea Time” discussions
 - on the news/progress on Web and IR industry/research
 - Apply **by the previous Thur. 13:00**
 - via email/WeChat for reservation
 - Bonus in the final score 
 - Examples:

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Introduction to WolframAlpha



What kind of queries can compute?



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Search Engine for Designers

- ▶ A meta search engine for multimedia contents
- ▶ A project by @ahmetsulek.
- ▶ The best royalty free stock photos are delivered now along with videos, mockups, icons, fonts and colors.

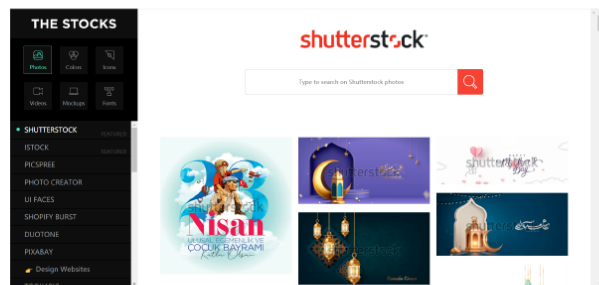


Figure 2: The Stocks

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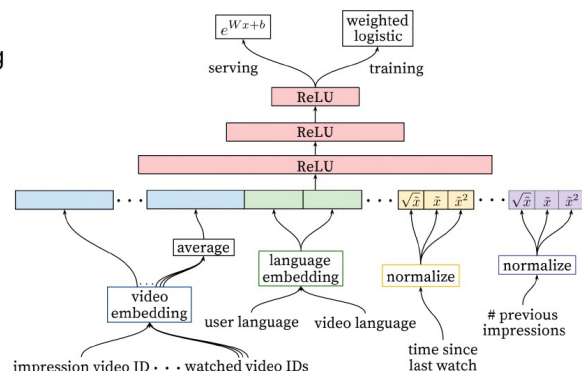
11

Youtube Recommendation Algorithm

YOUTUBE RECOMMENDATION ALGORITHM

Ranking Network

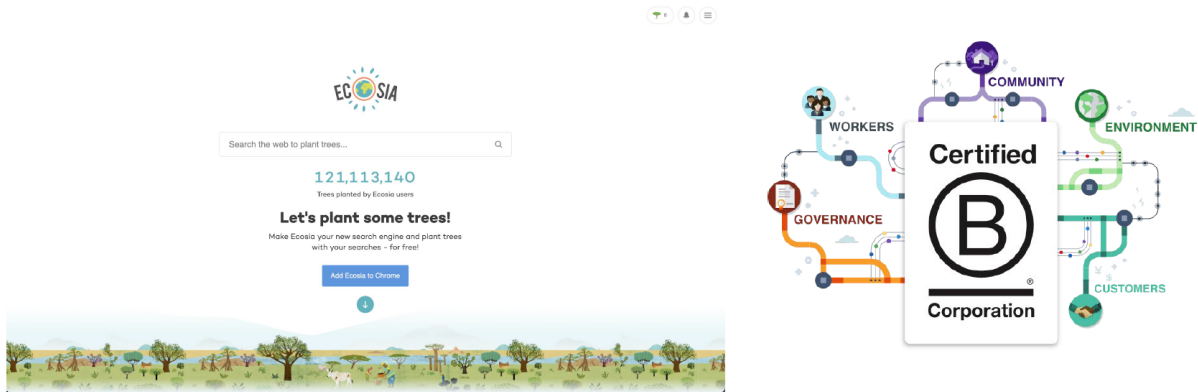
- The target is to represent a temporal sequence of user actions and how these actions relate to the video impression being scored.
- Categorical features are embedded whereas continuous features are normalized $[0,1]$ before concatenated into one large vector



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Ecosia: Transforming ad revenue to trees



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Recommendation in Pinterest

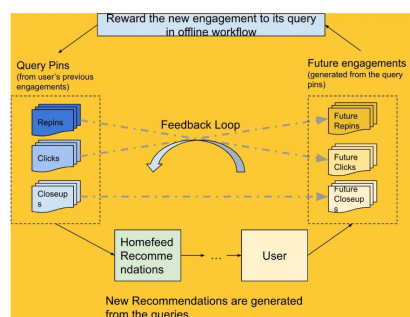
<https://medium.com/pinterest-engineering/pinnability-machine-learning-the-home-feed-64be2074bf60>

**Pinnability
(2015)**

<https://medium.com/pinterest-engineering/query-rewards-building-a-recommendation-feedback-loop-during-query-selection-70b4d20e5ea0>

**Query
Reward
(2022)**

~30% of recommended pins come from pin to pin-based retrieval.



Pinterest

Pinnability is the collective name of the machine learning models developed to help improve home feed.

Pinterest

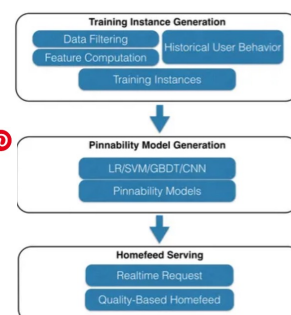
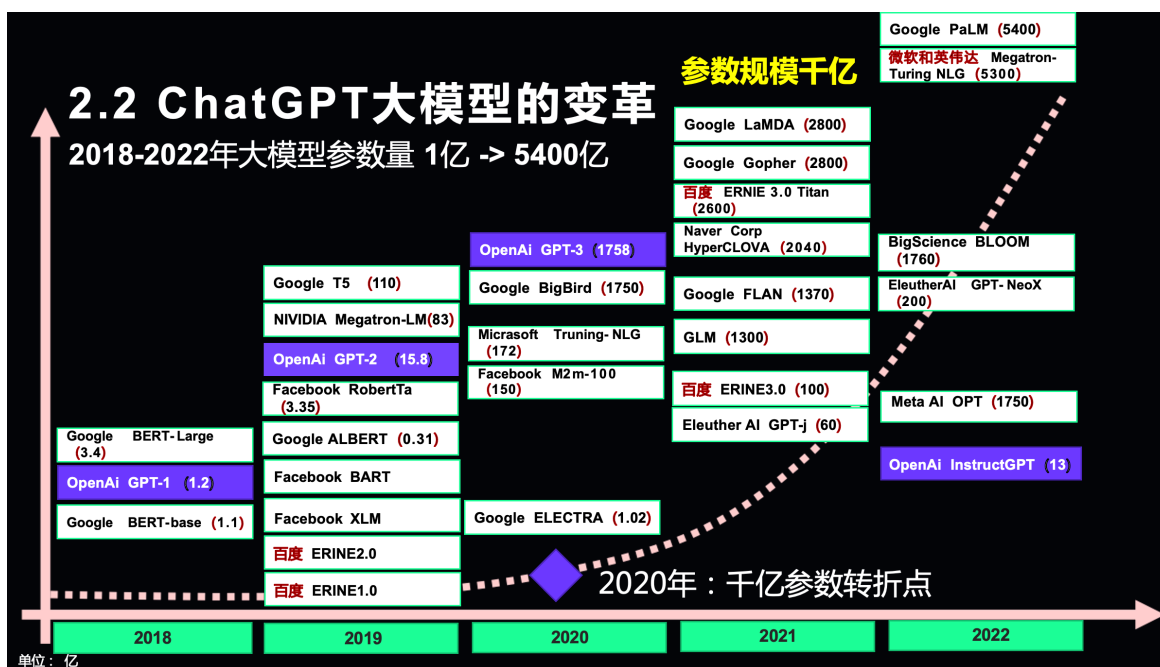


Fig 3: Pinnability pipeline overview.

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ChatGPT



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Course Arrangements (Tentative): Part II

■ Part II: weeks 9 &10, 12&13, Seminar by student groups

□ Everyone should join a group

■ 6~8 groups

□ A survey on new tech.

■ LLM and Search

■ LLM and Recommendation

■ LLM and User

■ Agentic IR

■ Multi-modal IR

■ New IR tasks

■ IR in specific scenarios

■

The Talk should include:

➤ Presentation:

➤ Background

➤ Approaches

➤ Organized into categories

➤ What are the differences?

➤ Pros and Cons

➤ Challenges

➤ Future Directions

➤ References

➤ QA

* Week 11: May 5th, National Holiday, NO classes, NO make-ups

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Course Arrangements (Tentative): Part III

- Part III: Week 14 -16, Course workshop
- 2 students as a **team**
- To **Design** or **Build** a [prototype Search Engine](#)
 - You [can use general IR toolkits](#) to build your search engine.
 - Both [general and domain-specific SEs](#) are acceptable.
 - A specific one with some interesting functions might be more attractive to your audiences
- **Presentation** on the course workshop: Talk + Demo + QA
- **Award** and celebration for “[the Best Project](#)”.

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Part III. Project and Course workshop (1)

- Two options: Option 1
 - 1. **Design Only**: [Design](#) the SE system [with detail info](#).
 - Models, equations, mathematical foundations, analysis on the characteristics and difficulties.....
 - **Easier but with less scores on general**
 - **Scoring: 60-85**
 - Longer QA time will be issued for Design Only talk
 - Metrics:

Presentation (1)	General Design (1)	Novelty (1)	Soundness of the Tech. (2.5)	QA (2)	Timing (1)	Total (8.5)
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Part III. Project and Course workshop (2)

- Two options: Option 2
 - ❑ 2. **Design & Implementation**: Design and then implement your prototype search engine
 - Harder one with generally higher scores
 - Scoring: 75-100, or 0.
 - The final score is based on talk, live demo, and QA
 - Metrics:

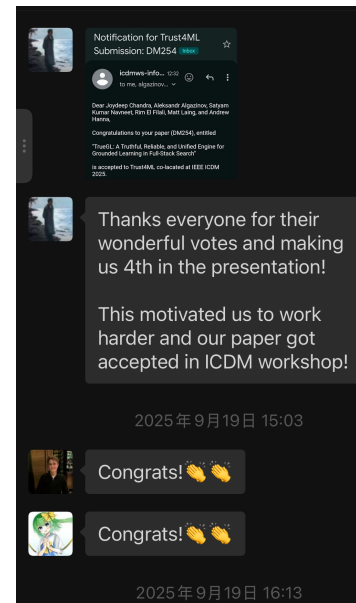
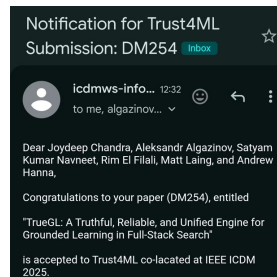
Presentation (1)	Soundness of the Tech. (2.5)	Pre-test (1)	Live Demo (2.5)	QA (2)	Timing (1)	Total (10)
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Part III. Project and Course workshop (3)

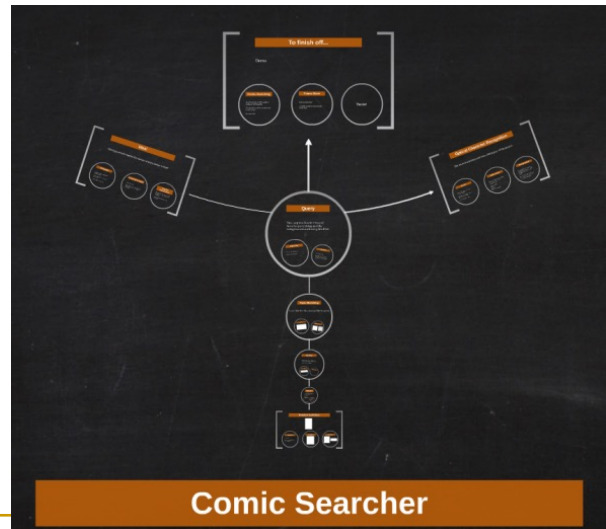
- Awards and celebration to “the Best Project”.



20

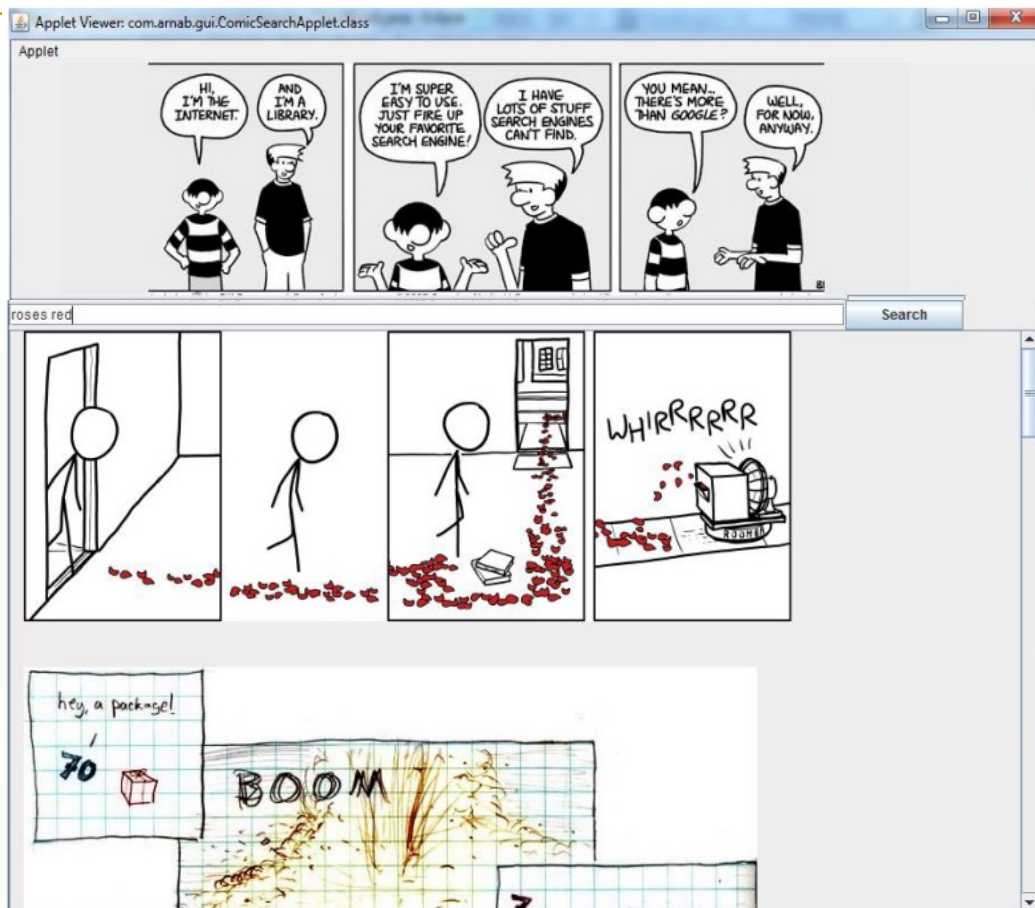
20

Examples of the past course projects



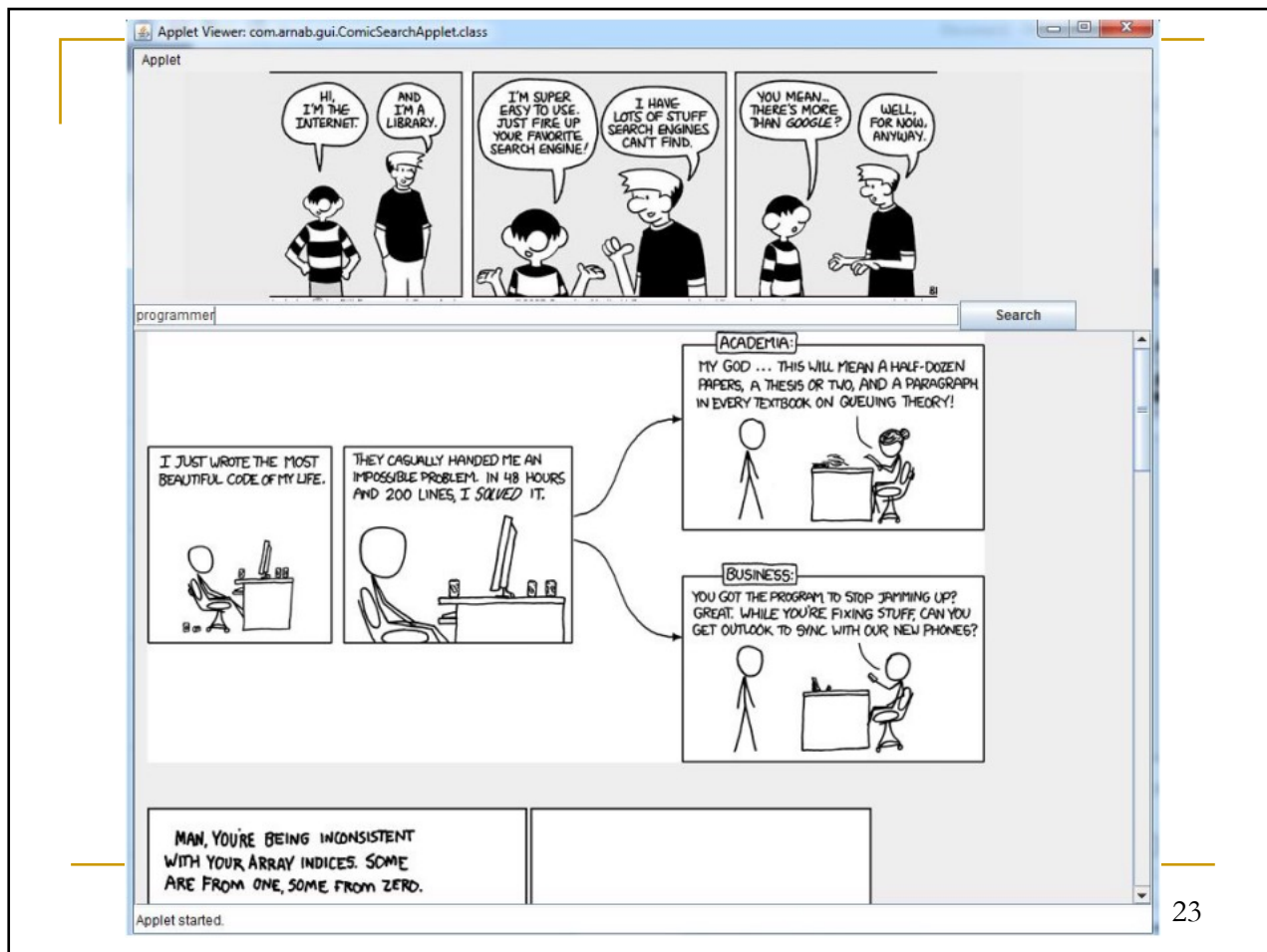
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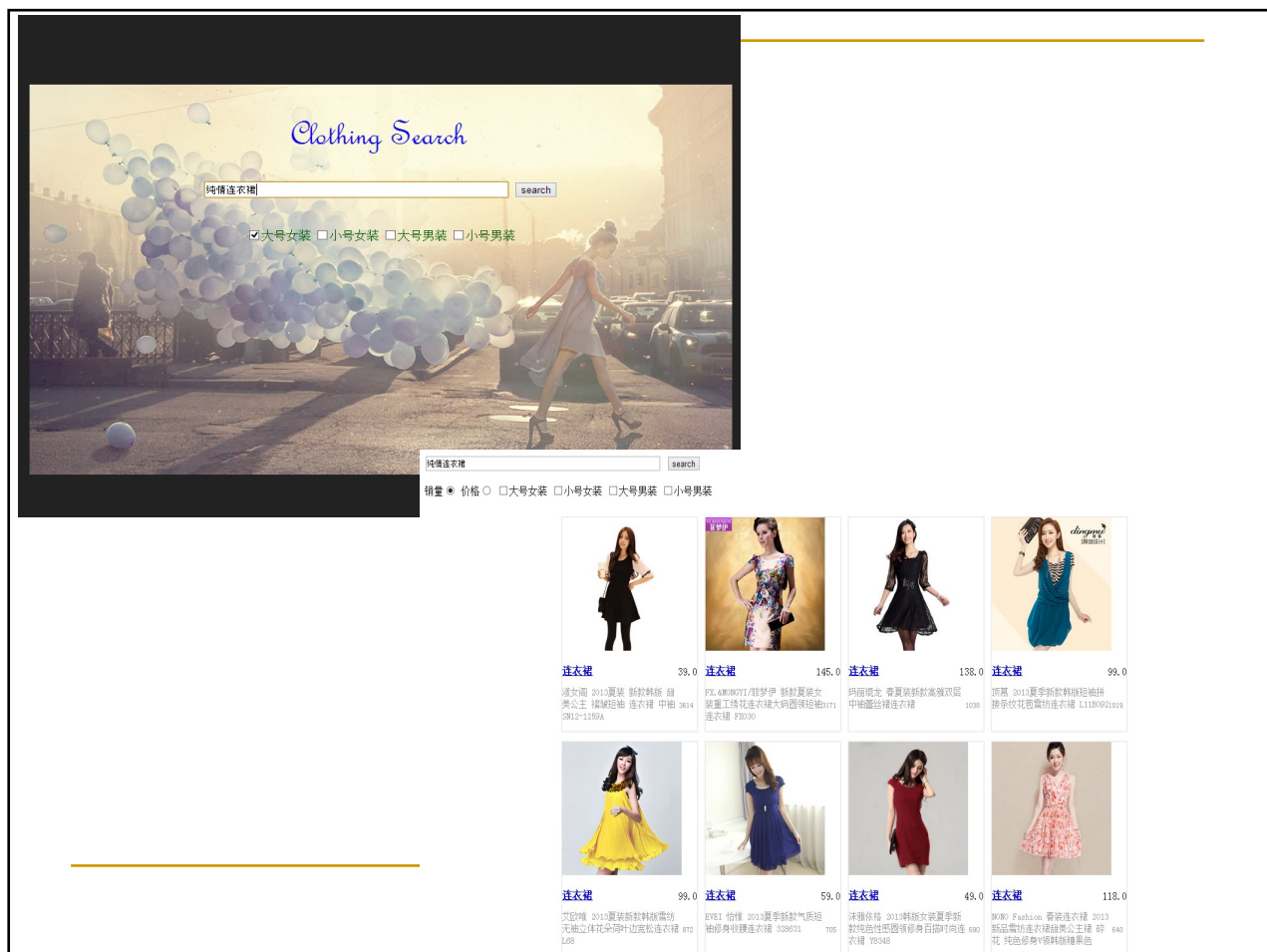
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Supports fuzzy search (maximum edit distance of 3)

L^AT_EX Symbol Search

esplon

Mouse over equations to display their T_EX commands

Symbol in T _E X	T _E X Command	Name
E, ϵ and ε	<code>E</code> , <code>\epsilon</code> and <code>\varepsilon</code>	

Can search content

L^AT_EX Symbol Search

the set of all real numbers

Mouse over equations to display their T_EX commands

Symbol in T _E X	T _E X Command	Name	Explanation
		Read as	
		Category	
$\mathbb{R}$	<code>\mathbb{R}</code>	real numbers	$\mathbb{R}$ means the set of real numbers.
$\mathbf{R}$	<code>\mathbf{R}</code>	R; the (set of) real numbers; the reals	
		numbers	

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theamazingemojisearcher.herokuapp.com/?q=happy

theamazingemojisearcher.herokuapp.com/emoji/3

The Amazing Emoji Searcher

Home / 🤔 face with tears of joy

Face with tears of joy



happy

sm

smi

Description

🤔 A laughing emoji which at small sizes is often mistaken for being tears of sadness. In fact, this emoji is laughing so much that it is crying tears of joy. Tears are coming from both eyes, not due to sadness, but overwhelmed with happiness or laughter.

Also known as

- 🤔 Laughing Emoji
- 🤔 Laughing Crying Emoji
- 🤔 Happy Tears Emoji
- 🤔 Laughing Tears Emoji
- 🤔 LOL Emoji

Different versions

iPhone

Android

Twitter

Gmail

Windows

Black and white

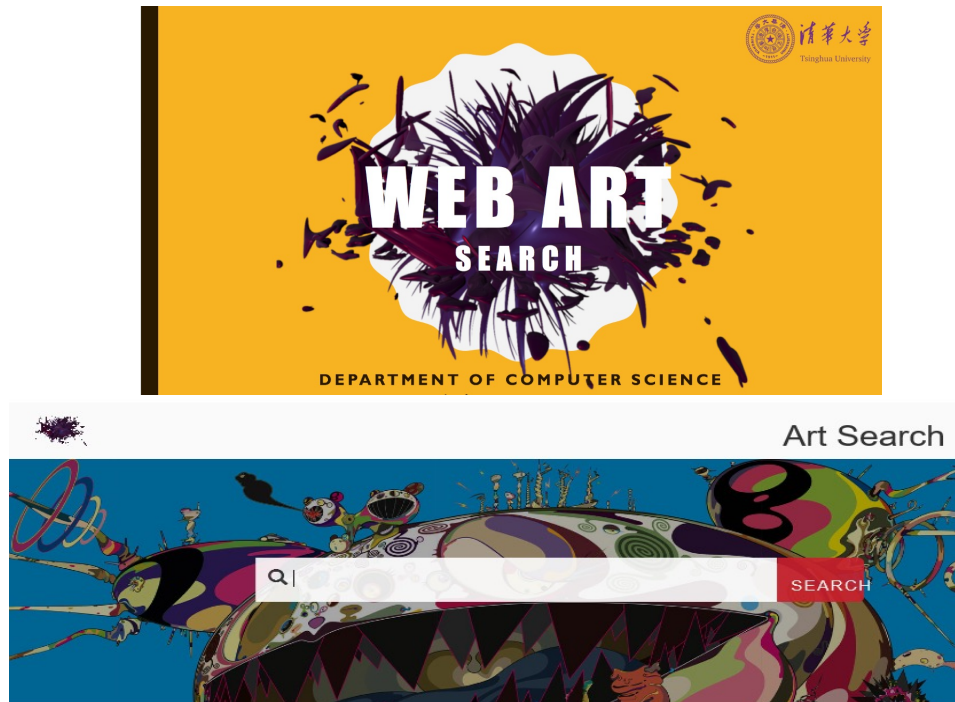
Related to

- 😊 Smiling Face With Open Mouth And Smiling Eyes
- 😭 Cat Face With Tears Of Joy
- 😭 Crying Face
- 😭 Loudly Crying Face
- 😊 Smiling Face With Open Mouth And Tightly-Closed Eyes

Made with love by Richard Luong Available on GitHub

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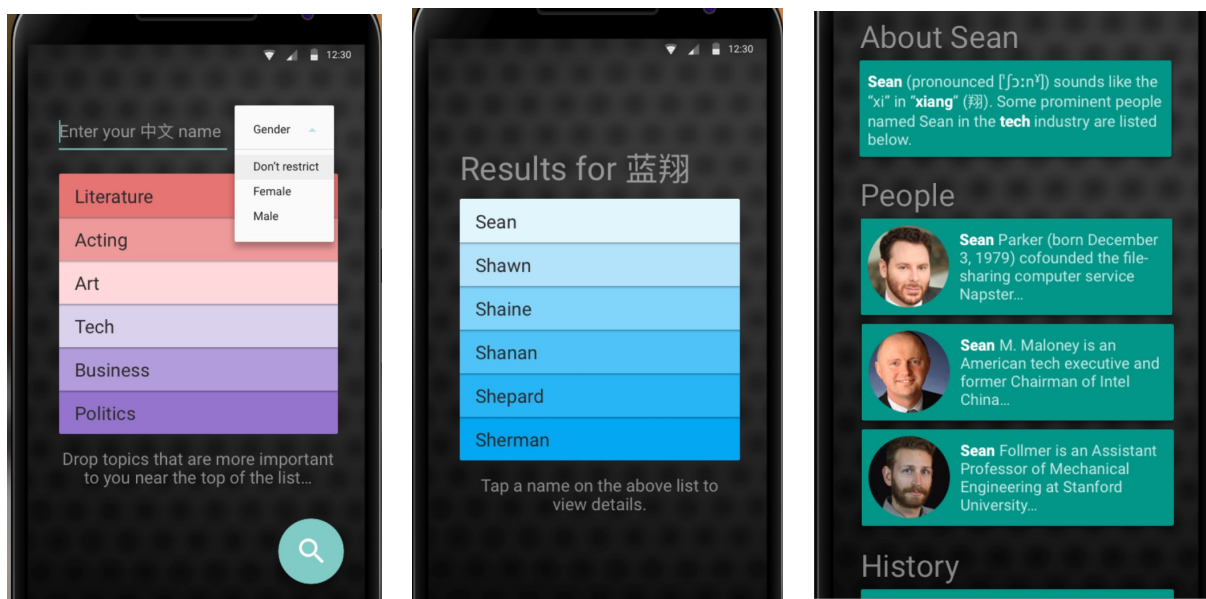
Web Art Search



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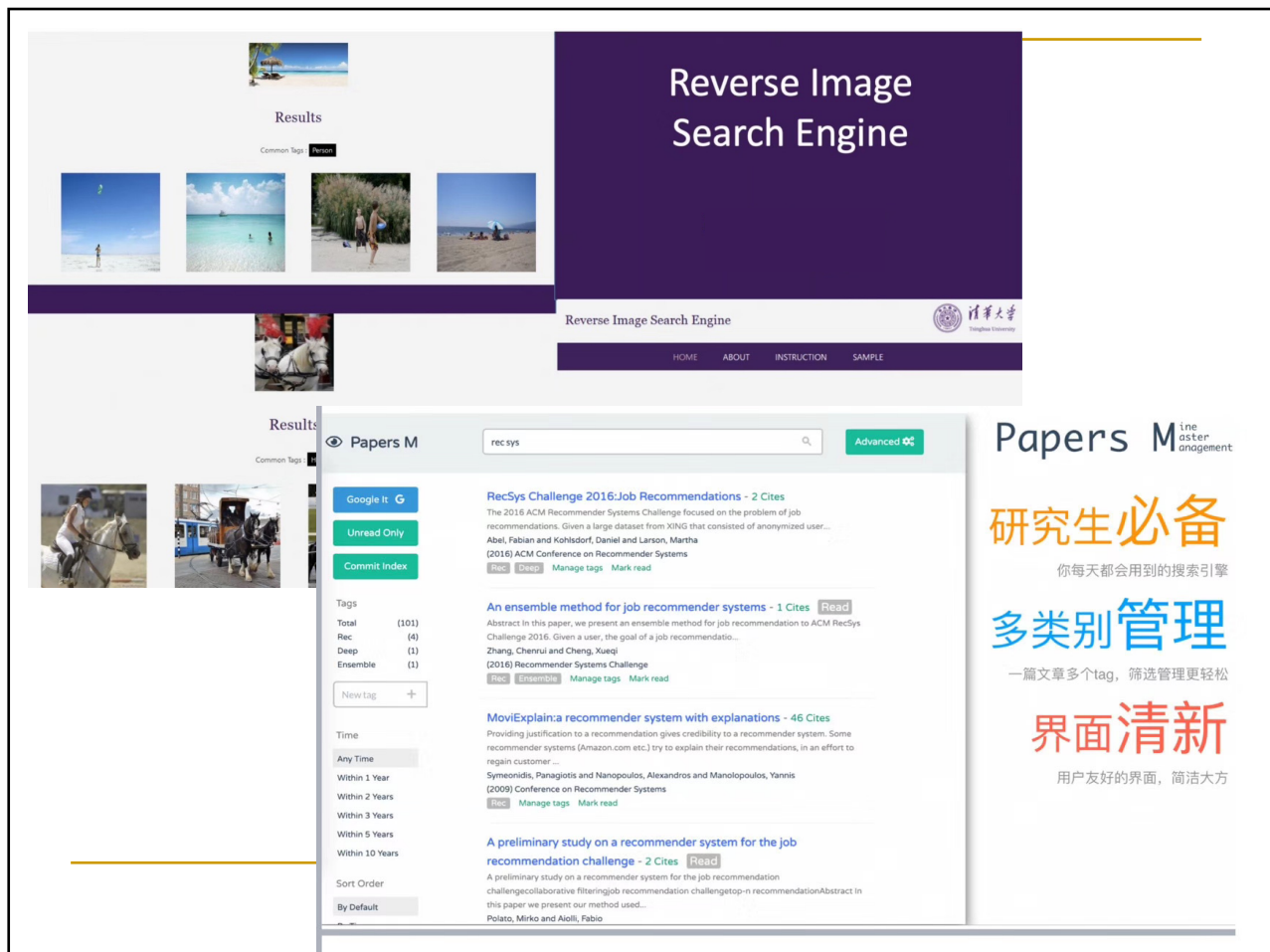
27

Bilingual Name Search Engine

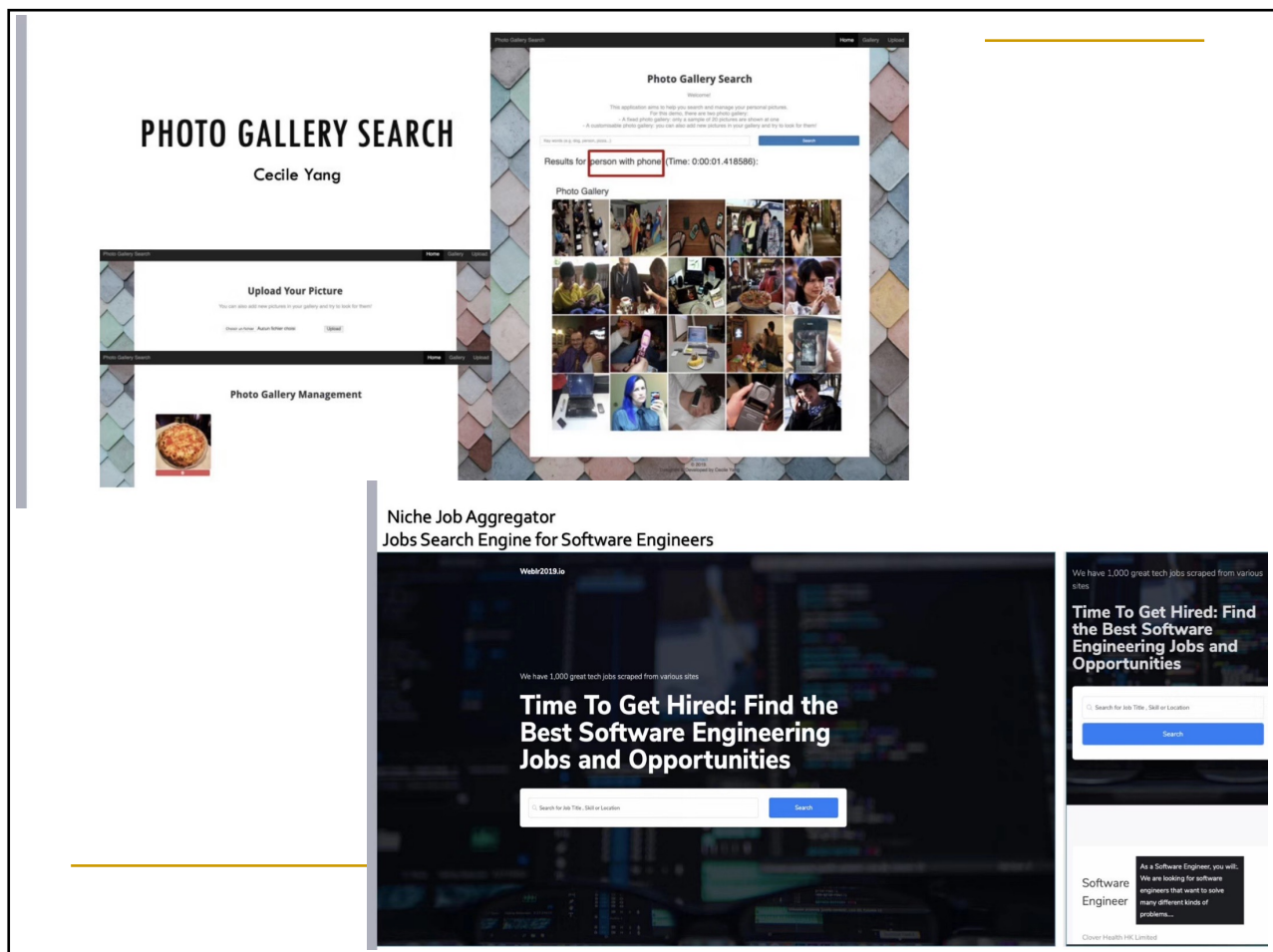


28

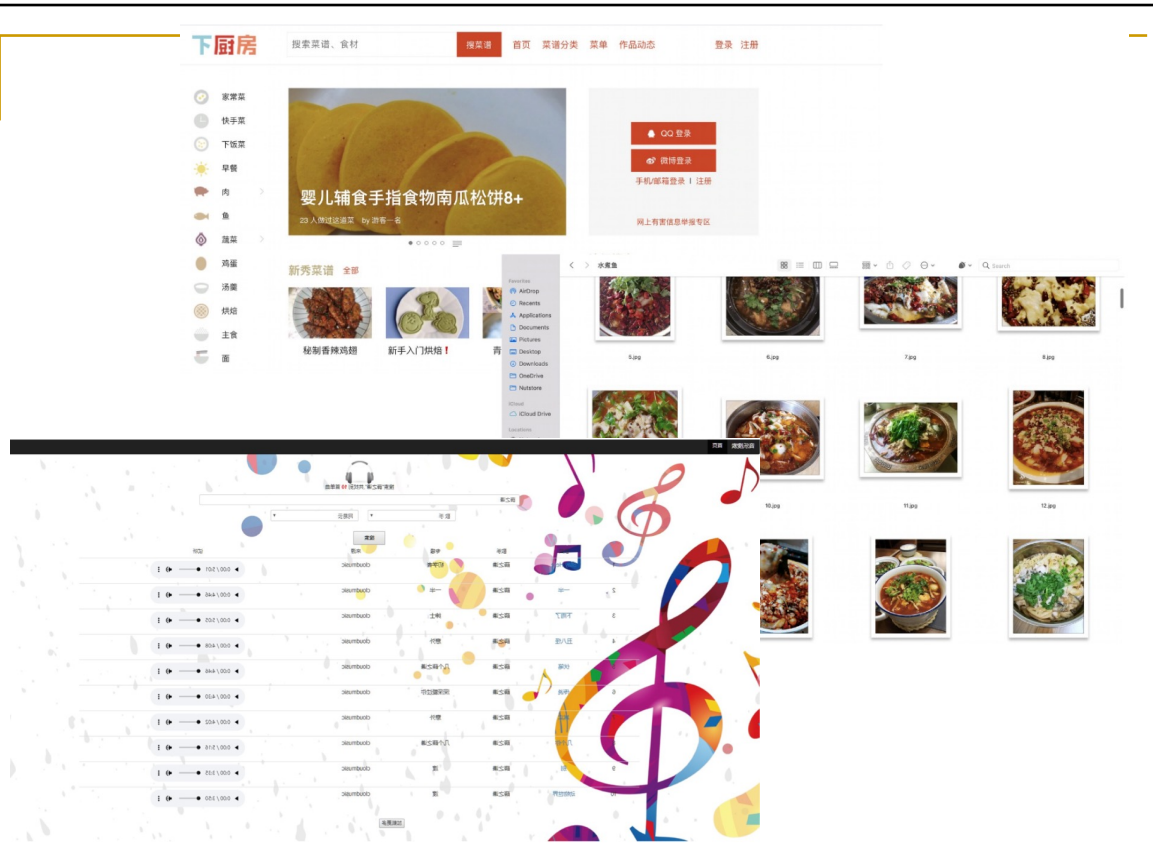
28



29



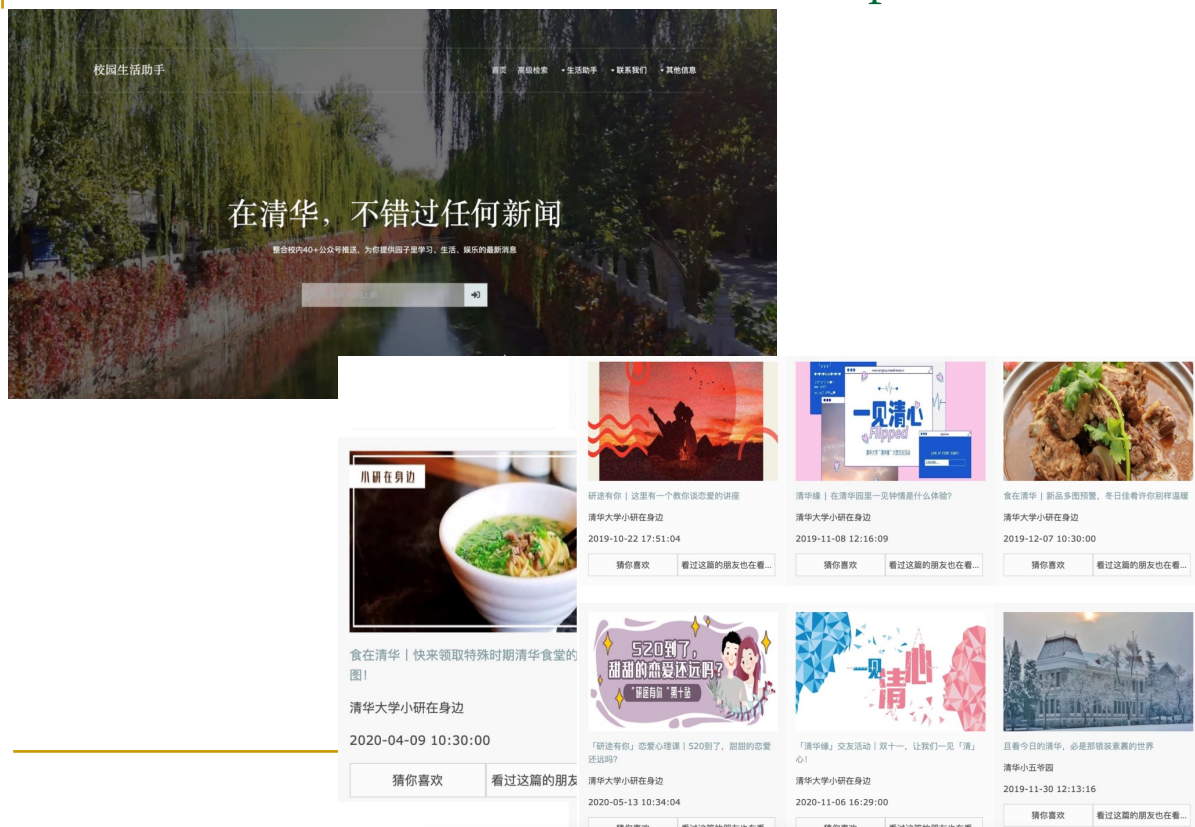
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Search and Recommendation for Campus Info.



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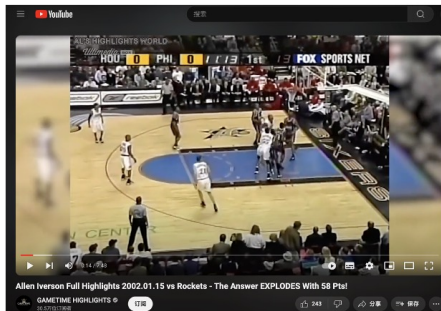
Sports Video Search Engine

球员检索：“阿伦 艾佛森”



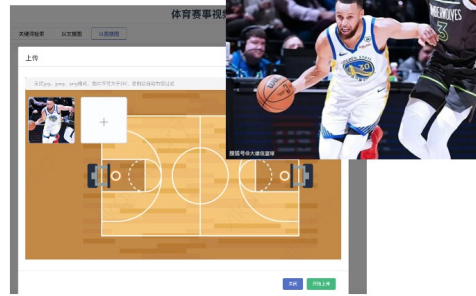
组合检索：

年份: 2001 + 主队: “76人” + 客队: “火箭”

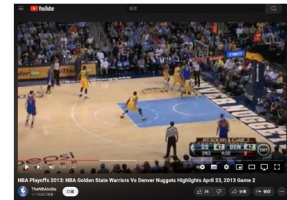


测试样例 IV 图片检索

检索I：斯蒂芬·库里



正确返回库里在勇士队的
相关比赛视频



俄州阿克伦，美国职业篮球

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Concerts Information SE

音乐演出检索

— 搜索你想要的演唱会、音乐节 —

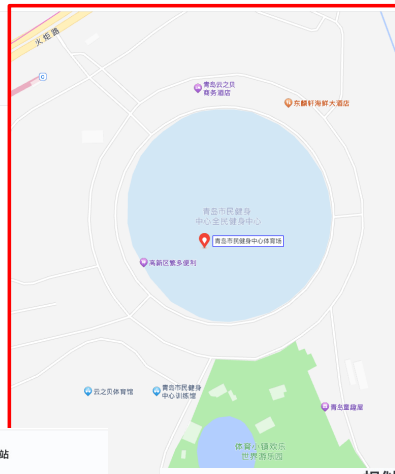
林俊杰

搜索

45 场相关的音乐演出

筛选

按相关度排序



地图定位

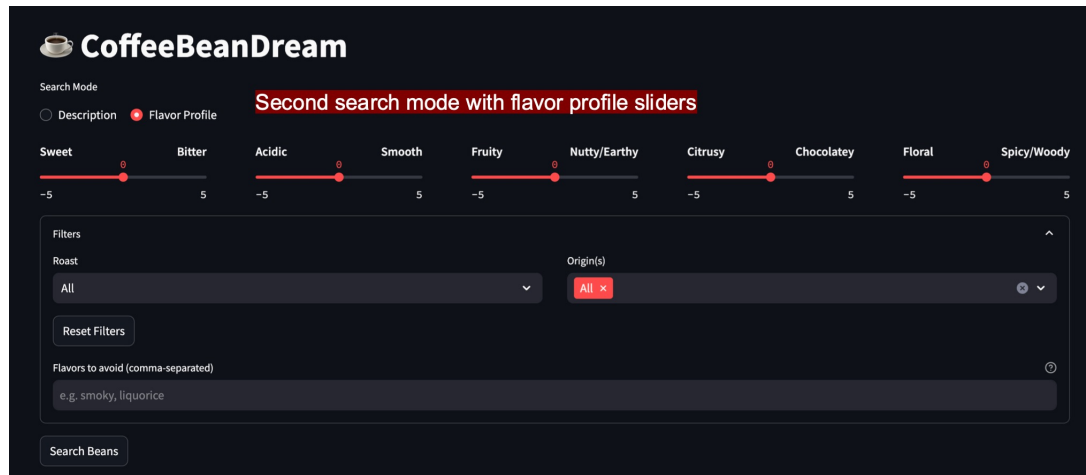
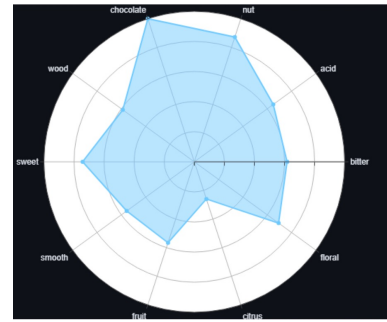
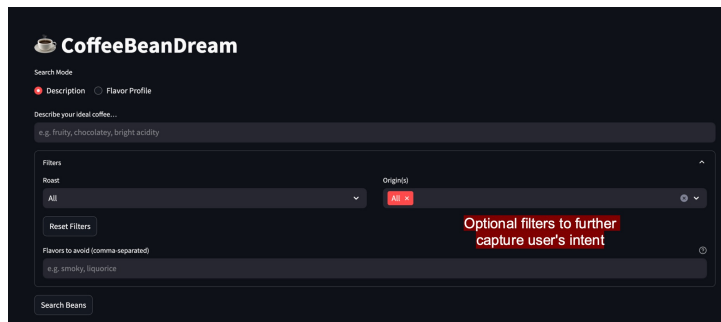
相似演出



地图跳转链接



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Search Engine
in Rhythm Game

Exam Search Engine

Difficulty:

✓ All

Easy

Intermediate

Advanced

Unknown

University:

Question Type:

Sort by:

Loading...

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Evaluation (Subject to modifications)

- Seminar (~30%)
- Workshop (60%), evaluated by
 - The other students (25 ~ 30%)
 - The teacher and TA (30% ~ 35%)
- QA and Course activities (~10%)
 - Activities in the seminar and workshop QA
- Bonus: 
 - Tea Time presentation
 - The best project in the workshop
 - More activities during the whole class

Active thinking and discussions are highly encouraged !

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References

- We're not having an official textbook
 - There isn't one with good coverage of all & only the topics we'll discuss
 - A **changing** field, **advanced** topics
- Some Books
 - Ryen W. White, Chirag Shah, **Information Access in the Era of Generative AI**
 - W. Bruce Croft, Donald Metzler, Trevor Strohman, **Search Engine: information retrieval in practice**
 - Christopher D. Manning, et al, **Introduction to information retrieval**
 - I. Witten, A. Moffat, and T. Bell, **Managing Gigabytes**
- Proceedings of Conferences
 - SIGIR, WWW, IJCAI, WSDM, RecSys, CIKM, TREC, NTCIR ...
- Very important: Web resources, Search engines, **LLMs (need verifications)**

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I. IR introduction

– 1. What's IR?



Figure Copyright by TREC

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What's IR? (cont.)

■ Goal:

- Find documents *relevant* to *an information need* from a large *document set*

■ And now:

- *Beyond relevance*
 - Timeliness, diversity, quality, fairness, privacy
- *Multi-modal documents*
- *Users' (implicit) information need* and *new UI*
- *Heterogeneous environment*



Figure Copyright by TREC

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I. IR introduction

– 2. IR history



Figure Copyright by TREC

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History of IR* on One Slide

- Birth of IR **(The history of Web Search will be discussed in later lectures)*
 - 1945: Vannevar Bush's article "As we may think"
 - 1957: H. P. Luhn's idea of word counting and matching

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As we may think



This article is reprinted in its entirety, with permission, from The Atlantic Monthly, July, 1945. A condensation was printed by Life Magazine in 1945, with illustrations. The article has been reprinted variously since then; it can be found at The Atlantic's own site, at <http://www2.theatlantic.com/atlantic/atlweb/fushbbs/compuser/tech.htm> and also at <http://www.iq.sfu.ca/~duscher/mic/vbush/>.

As We May Think Vannevar Bush

As Director of the Office of Scientific Research and Development, Dr. Vannevar Bush has coordinated the activities of some six thousand leading American scientists in the application of science to warfare. In this significant article he holds up an incentive for scientists when the fighting has ceased. He urges that men of science should then turn to the massive task of making more accessible our bewildering store of knowledge. For many years inventions have extended man's physical powers rather than the powers of his mind. Trip hammers that multiply the fists, microscopes that sharpen the eye, and engines of destruction and detection are new results, but not the end results, of modern science. Now, says Dr. Bush, instruments are at hand which, if properly developed, will give man access to and command over the inherited knowledge of the ages. The perfection of these pacific instruments should be the first objective of our scientists as they emerge from their war work. Like Emerson's famous address of 1837 on "The American Scholar," this paper by Dr. Bush calls for a new relationship between thinking man and the sum of our knowledge.

—The [Atlantic Monthly] Editor, July 1945

INTRODUCTION . . . MARCH 1956



Set a **goal** of fast access
to the contents of the
world's libraries:

- A **1M** book library

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LUHN H.P.

- LUHN, H.P., 'A statistical approach to mechanised encoding and searching of library information', *IBM Journal of Research and Development*, 1, 309-317 (1957).
- 'It is here proposed that the frequency of word occurrence in an article furnishes a useful measurement of word significance. It is further proposed that the relative position within a sentence of words having given values of significance furnish a useful measurement for determining the significance of sentences. The significance factor of a sentence will therefore be based on a combination of these two measurements.'

---- LUHN, H.P., 'The automatic creation of literature abstracts', *IBM Journal of Research and Development*, 2, 159-165 (1958).

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History of IR* on One Slide

- Birth of IR **(The history of Web Search will be discussed in later lectures)*
 - 1945: Vannevar Bush's article "As we may think"
 - 1957: H. P. Luhn's idea of word counting and matching
- Indexing & Evaluation Methodology (1960's)
 - Smart system (G. Salton's group)
 - Cranfield test collection (C. Cleverdon's group)
 - Indexing: automatic can be as good as manual (controlled vocabulary)
- IR Models (1970's & 1980's, late 1990's & early 2000's, 2009~2015) ...
- Large-scale Evaluation & Applications (1990's)
 - TREC (D. Harman & E. Voorhees, NIST), CLEF, NTCIR, ...
- Web search (2000's): Search engines, Boundary are disappearing
- Vertical Search, Knowledge, Social, User (2010's)
- Dense retrieval, Conversational IR, LLM and RAG (2020's ~)

(Retrieve Augmented Generation)

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